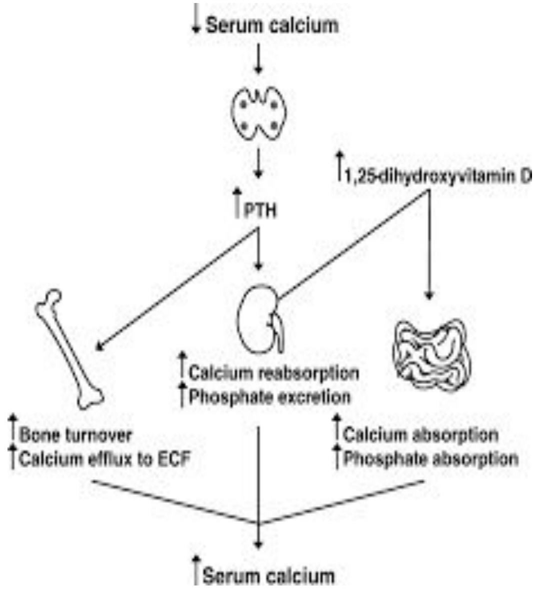
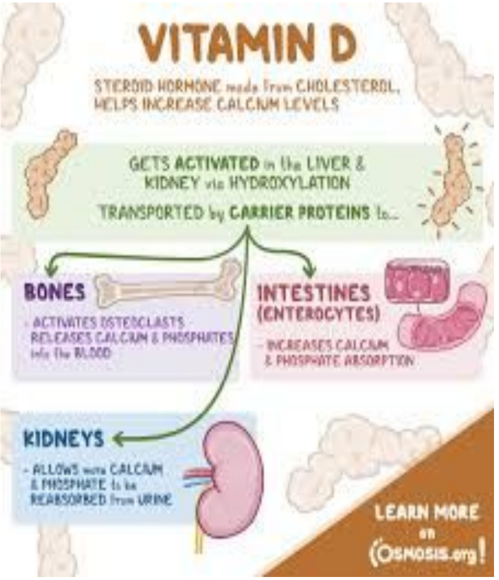
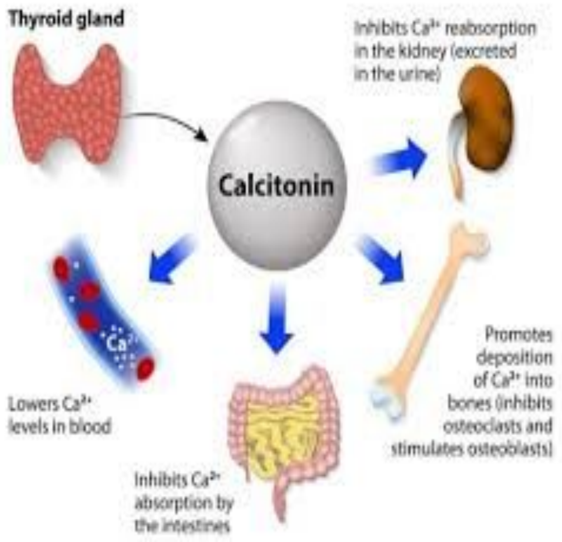
CALCIUM-PHOSPHATE METABOLISM: PARATHYROID HORMONE, VITAMIN D, CALCITONIN







Dr. Humaira Aman Ali

Objectives

- At the end of the lecture, students would be able to:
    - Know the body distribution of calcium and phosphate.
    - Describe the sources and functions of calcium and phosphate.
    - Explain the causes of hyper and hypo calcemia and phosphatemia.
    - Understand the role of parathyroid hormone, vitamin D and calcitonin in the regulation of calcium and phosphate metabolism.


CALCIUM
Ca
Ca
Ca
Ca
Ca

CALCIUM (Ca)

- Calcium is an important mineral mainly found in bone and teeth.

Body distribution:

- Calcium forms about 1% to 1.5% of an average adult's body weight.
- 99% of which is seen in bones and teeth as carbonate or phosphate of calcium.
- 1% of total body Ca occurs in extracellular fluid and soft tissues.

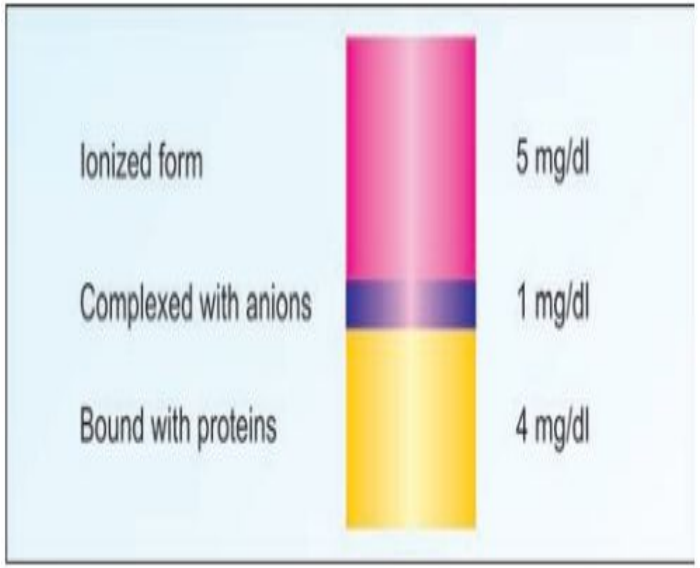
Different forms of calcium in PLASMA

- The normal level of plasma calcium is 9-11 mg/dl.
- The calcium in plasma is of 2 forms:

  - Non diffusible or Plasma protein bound calcium. Albumin is the major protein with which calcium is bound.

  - Diffusible:

1. Ionised calcium($Ca2+$): 40% of total calcium is in ionised form and is physiologically active form.
2. Complexed calcium, it is probably complexed with plasma anions such as citrate and phosphate.



Dietary sources of Calcium:

- Milk and Yoghurt is a good source for calcium
- Cheese
- Egg-yolk
- Beans
- Lentils
- Nuts. Of all nuts, almonds are among the highest in calcium. Figs
- Fish and vegetables are medium sources for calcium
- Cereals (wheat, rice) contain only small amount of calcium.

Contents of Daily Calcium Requirements by Age and Sex

Age and Gender	Adequate Daily Intake of Calcium (mg)
Infants, male or female, 0–6 months	210
Infants, male or female, 7–12 months	270
Children, male or female, 1–3 years	500
Children, male or female, 4–8 years	800
Males and females, 9–18 years	1300
Males and females,* 19–50 years	1000
Males and females,† 51 years and older	1200
Pregnant or lactating female, 14–18 years	1300
Pregnant or lactating female, 19–50 years	1000

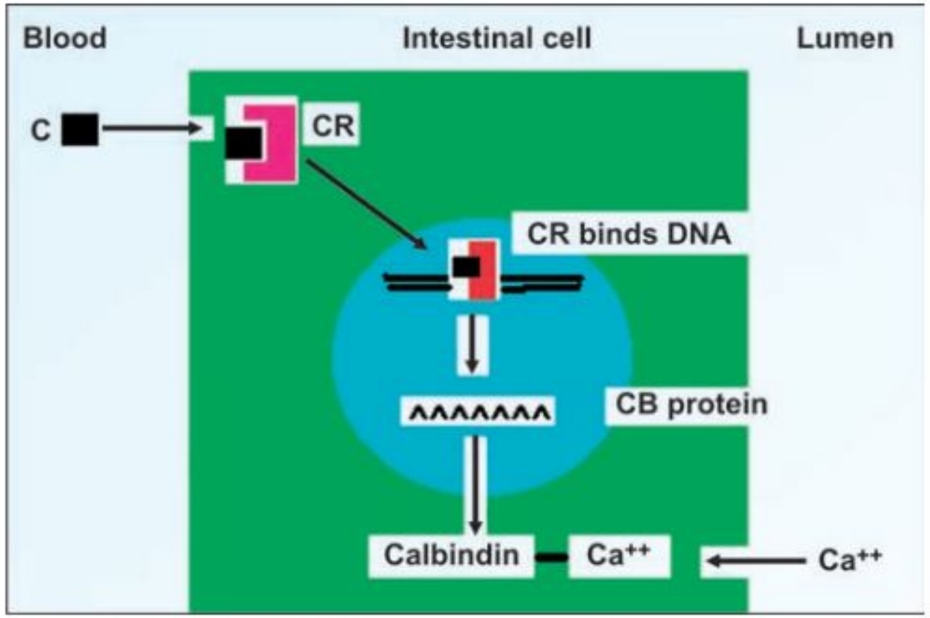
https://cdn.citl.illinois.edu/courses/KIN122/week6/chapter9/web_data/file10contents.htm

Absorption

- Calcium is absorbed mainly from the duodenum and first half of jejunum.

MECHANISM:

- Two mechanisms have been proposed for absorption of calcium by gut mucosa.
  - 1. Simple diffusion
  - 2. Calcium is absorbed against a concentration gradient, an "active" transport process requires energy and Ca++ pump.
  - Both the processes require 1, 25-di hydroxy-D3 (calcitriol).



Calcitriol increases calcium absorption. C = calcitriol; CR = calcitriol receptor complex; CB calbindin

FACTORS AFFECTING ABSORPTION

- **Factors causing increased absorption:**
  - ✔ Vitamin D: Calcitriol induces the synthesis of the carrier protein (Calbindin) in the intestinal epithelial cells, and so facilitates the absorption of calcium.
  - ✔ Parathyroid hormone: It increases calcium transport from the intestinal cells.
  - ✔ An acidic pH favours calcium absorption because the Ca-salts, particularly PO4 and carbonates are quite soluble in acid solutions.
  - ✔ Amino acids: Lysine and arginine increase calcium absorption

- **Factors causing decreased absorption:**

  - ✔ Phytic acid: found in plant seeds, prevents the absorption of iron, zinc, and calcium and may promote mineral deficiencies.

  - ✔ Malabsorption syndromes: Fatty acid is not absorbed, causing formation of insoluble calcium salt of fatty acid.

  - ✔ Oxalates and phosphates

FUNCTIONS

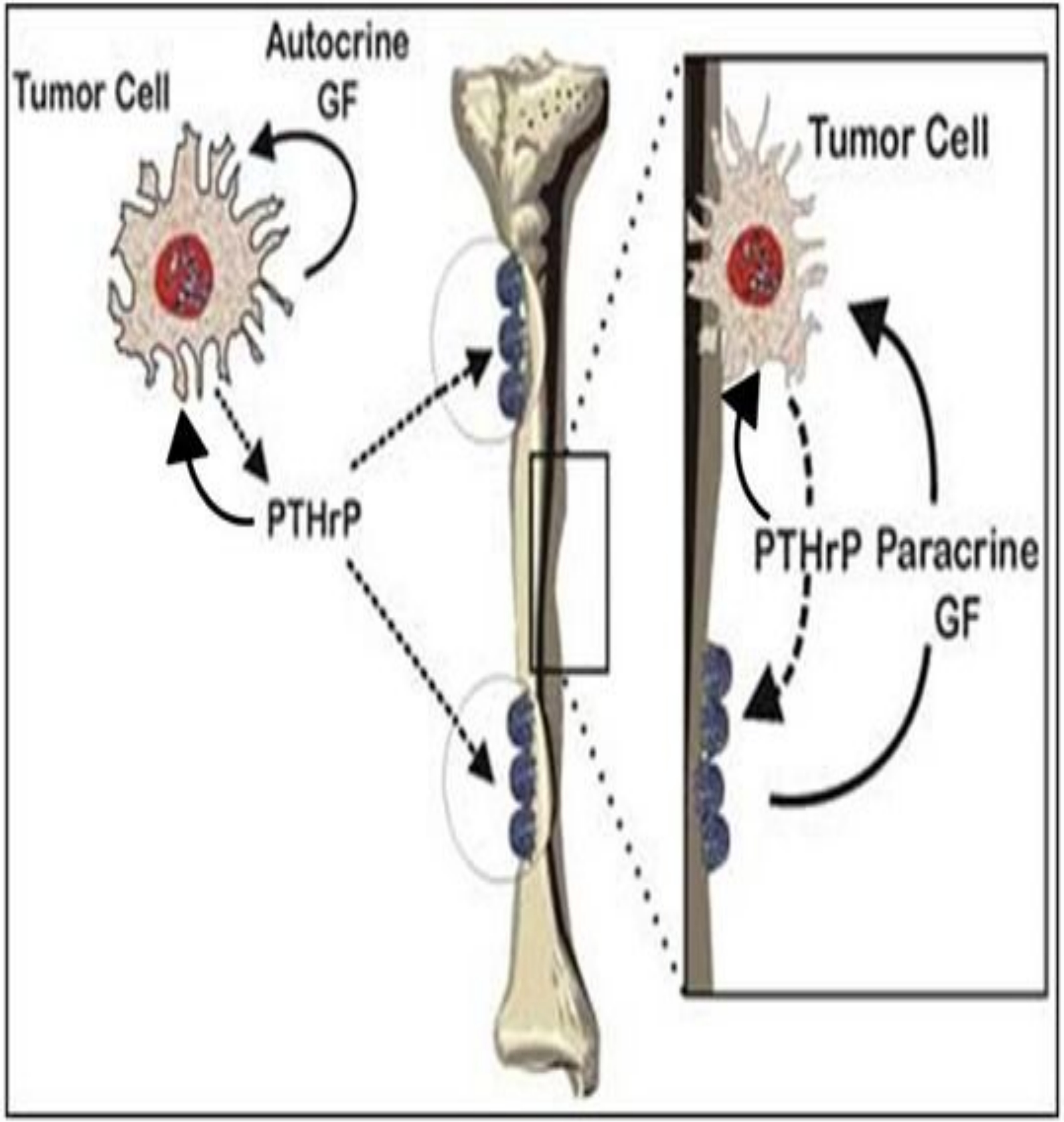
- Calcification of Bones and Teeth.
- Calcium plays a role in blood coagulation: Calcium is known as factor IV in blood coagulation cascade.
- Nerve conduction: Calcium is necessary for transmission of nerve impulses from presynaptic to postsynaptic region .
- Calcium mediates excitation and contraction of muscle fibers.
- Myocardium: Ca^{++} prolongs systole. In hypercalcemia, cardiac arrest is seen in systole. This fact should be kept in mind when calcium is administered intravenously. It should be given very slowly.
- Secretion of hormones: Calcium mediates secretion of insulin, parathyroid hormone, calcitonin, vasopressin, etc. from the cells.
- Activation of enzymes.

Hypercalcaemia

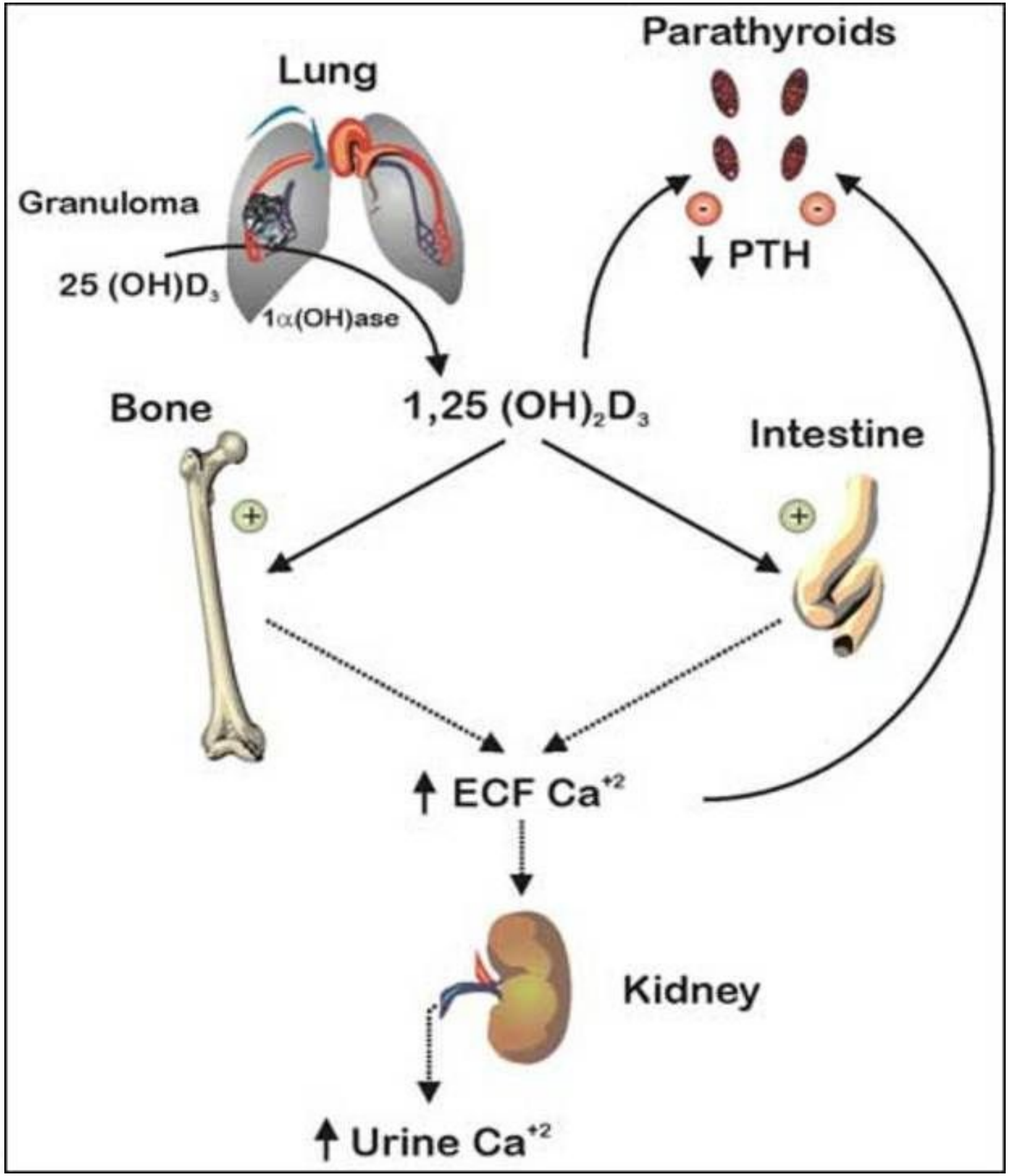
- When the serum calcium level exceeds **11.0 mg/dl,** it is called as hypercalcaemia.

Causes:

1. **Primary hyperparathyroidism:** Most common cause
    - Familial
    - Tumours – Solitary adenoma, Multiple adenomas, Parathyroid carcinoma
2. **“Ectopic” hyperparathyroidism:** Multiple endocrine neoplasia type I, II (MEN I) & (MEN II)
3. **Malignancy:**
    - No direct skeletal involvement:
        - PTH related protein (PTHrP)
    - Metastatic carcinoma of bone – Direct erosion of bone by tumour
4. **Haematological malignancies:** Multiple myeloma



Tumor Cell
Autocrine GF
PTHrP
Tumor Cell
PTHrP
Paracrine GF



Parathyroids
Lung
Granuloma
25 (OH)D_3
1α(OH)ase
↓ PTH
Bone
1,25 $(OH)_2D_3$
Intestine
↑ ECF Ca^{+2}
Kidney
↑ Urine Ca^{+2}

cont..

5. **Other Endocrine Causes:** Hyperthyroidism (Thyrotoxicosis), Acute adrenal insufficiency (Addison's disease)

6. **Granulomatous Diseases:** Tuberculosis and sarcoidosis etc..

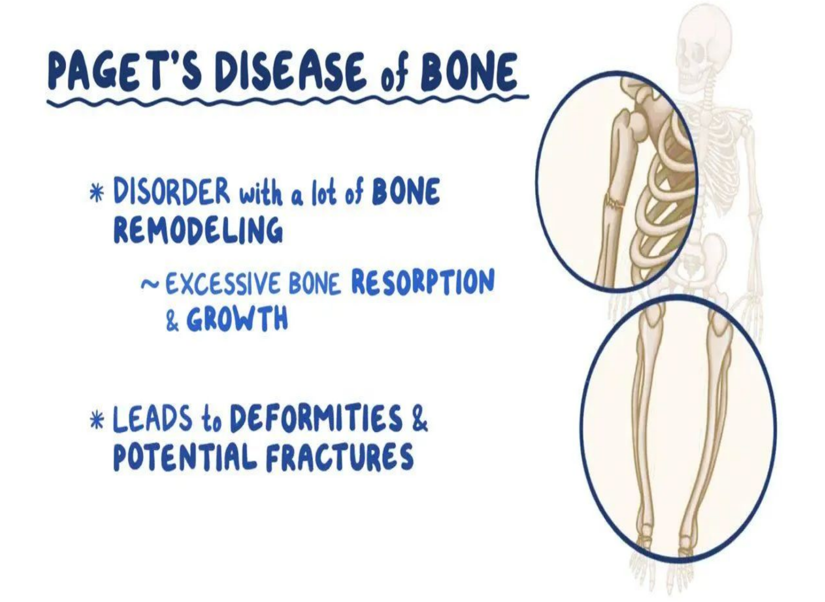
7. **Overdosage of Vitamins:** Vitamin A intoxication, and Hypervitaminosis D

8. **Drug-induced Hypercalcaemia (Iatrogenic):** Thiazide diuretics (can increase renal tubular reabsorption of calcium)

cont..

9. Miscellaneous Other Causes:

- Idiopathic hypercalcemia of infancy (William syndrome)
- Paget's disease of bone: a chronic disease characterized by episodic accelerated bone resorption and growth of abnormal replacement bone, causing bone pain and deformation
- Prolonged immobilization because of rapid bone turnover
- Milk-alkali syndrome: caused by the ingestion of large amounts of calcium supplements.



Symptoms of Hypercalcemia

1. Anorexia, nausea, vomiting
2. Polyuria and polydipsia (reduced concentrating ability)
3. Confusion, depression, psychosis and coma which can be fatal
4. Osteoporosis and pathological fracture
5. Renal stones
6. Ectopic calcification and pancreatitis
7. Abnormal heart rhythm (arrhythmia) with shortening of the QT interval.

Hypocalcaemia

- Hypocalcaemia is said to exist when serum calcium is less than 8.5 mg/dl.

Causes:

1. **Hypoalbuminemia:**
   - Reduction in serum albumin: Malnutrition, malabsorption states, nephrotic syndrome, chronic liver disease and liver failure, protein losing enteropathy

2. **Hypoparathyroidism:**
   - Surgical-induced partial or complete hypoparathyroidism (90% cases)
   - Idiopathic—may be autoimmune (10% cases)

3. **Renal Diseases and Renal Failure:**
   - Acute tubular necrosis
   - Chronic renal failure: – b/c of Impaired synthesis of 1,25-diOH D3 (calcitriol)

4. **Deficiency of Vitamin D:**

- Decreased exposure to sunlight
- Malabsorption
- Dietary deficiency
- Hepatic diseases
- Decreased renal synthesis of calcitriol

5. **Increased Calcitonin:**

- Medullary carcinoma of thyroid

6. **Increase in Phosphorus Level**

- Renal failure
- Phosphate infusion

Symptoms of Hypocalcemia

- Muscle cramps
- Paresthesia, especially in fingers
- Neuromuscular irritability, muscle twitchings
- Tetany
- Seizures
- Bradycardia
- Prolonged QT interval

Hypocalcemia causes increased neuromuscular excitability by decreasing the threshold needed for the activation of neurons. As a result, neurons become unstable and fire spontaneous action potentials that trigger the involuntary contraction of the muscles, which eventually leads to tetany.

Tetany

- Tetany is the occurrence of intermittent spasms, or involuntary contractions, of muscles, particularly in the upper and lower limbs and in the larynx.

❖ **Manifest tetany:** spontaneous clinical manifestations

- Classic signs of peripheral hyperexcitability of motor nerve: spasm of the muscles of the wrists and ankles (carpopedal spasm) and of the vocal cords (laryngospasm)
- Sensory manifestations: paresthesia, numbness, and tingling of the hands and feet
- Motor excitability of the central nervous system: brief, recurrent convulsions; usually generalized but may be localized to one side of the body

- **Latent tetany:** clinical signs not evident but can be elicited after stimulation.
- Chvostek's sign: (described by Frantisek Chvostek physician in 1876)
    - Tapping over facial nerve causes facial contraction will be positive
- Trousseu's sign: (French physician Armand Trousseau described it in 1861)
    - Trousseau's sign for latent tetany is most commonly positive in the setting of hypocalcemia.
    - The sign is observable as a carpopedal spasm induced by ischemia secondary to the inflation of a sphygmomanometer cuff, commonly on an individual's arm, to 20 mmHg over their systolic blood pressure for 3 minutes.
    - The carpopedal spasm is visualized as flexion of the wrist, thumb, and metacarpophalangeal joints with hyperextension of the fingers.

PHOSPHORUS

- Phosphate plays a crucial role in cell structure, signaling, and metabolism.
- Phosphate is found both as mineral phosphate(inorganic phosphate) and organic phosphate (phosphoric esters).

- **Food Sources:** Foods rich in phosphorus content are cheese, milk, nuts, organ meats, egg.

- **Body Distribution:**
  - Total body phosphate is about 500-700 g.
  - 85 % is found in bones and teeth
  - 14% in soft tissues
  - 1% is found in ECF, less than 1% circulating in serum
  - Phosphate is mainly an intracellular ion and is seen in all cells.

- **Daily requirement:** About 1 g of phosphate is required to be taken in the diet daily.

Absorption:

- 90 per cent of daily dietary phosphate is absorbed.
- The absorption is stimulated by both PTH and Vit. D3.
- Phosphorus holds an inverse relationship with calcium. An excess of serum calcium or phosphate results in the excretion of the other by the kidney.
- PTH increases calcium and phosphate release from the bone and decreases loss of calcium and increases loss of phosphate in the urine.

Functions of Phosphate Ions

1. Formation of bone and teeth.
2. Production of high energy phosphate compounds such as ATP, CTP, GTP, creatine phosphate, etc.
3. Synthesis of nucleoside co-enzymes such as NAD and NADP.
4. DNA and RNA synthesis, where phosphodiester linkages form the backbone of the structure.
5. Formation of phosphate esters such as glucose-6-phosphate, phospholipids.
6. Formation of phosphoproteins, e.g. casein.
7. Activation of enzymes by phosphorylation.
8. Phosphate buffer system in blood. The ratio of

Causes of Hyperphosphatemia

1. Increased absorption of phosphate

Excess vitamin D

Phosphate infusion

2. Increased cell lysis

Chemotherapy for cancer

Bone secondaries

Rhabdomyolysis

3. Decreased excretion of phosphorus

Renal impairment

Hypoparathyroidism

4. Hypocalcemia

5. Massive blood transfusions

6. Thyrotoxicosis

7. Drugs

Causes of Hypophosphatemia

Decreased absorption of phosphate
- Malnutrition
- Malabsorption
- Chronic diarrhea
- Vitamin D deficiency

Intracellular shift
- Insulin therapy, glucose phosphorylation
- Respiratory alkalosis

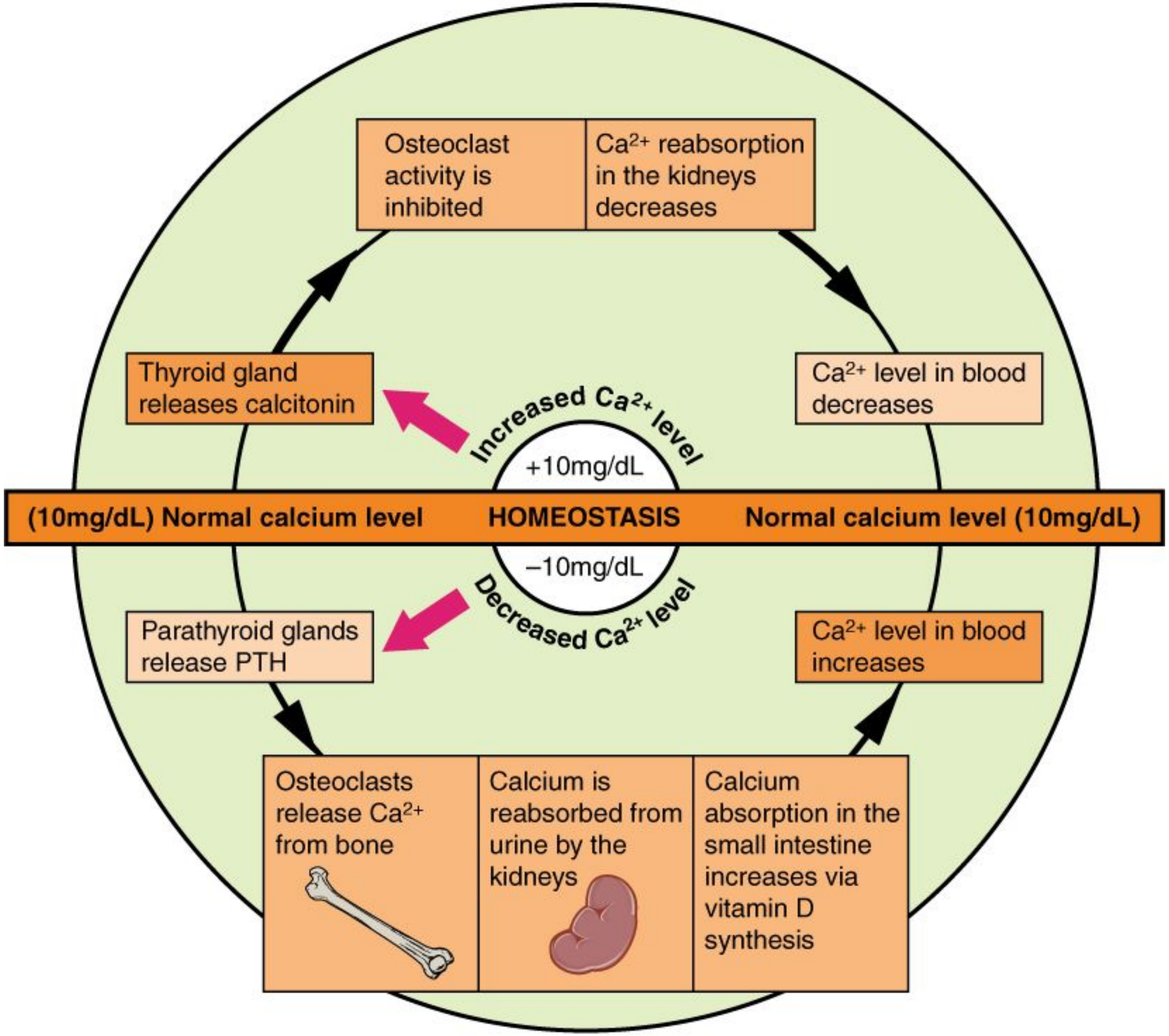
Increased urinary excretion of phosphate
- Hyperparathyroidism

Hypercalcemia

Chronic alcoholism

Drugs
- Antacids, Diuretics, Salicylate intoxication

REGULATION


Osteoclast activity is inhibited
Ca2+ reabsorption in the kidneys decreases
Ca2+ level in blood decreases
Thyroid gland releases calcitonin
Increased Ca2+ level
+10mg/dL
(10mg/dL) Normal calcium level
HOMEOSTASIS
Normal calcium level (10mg/dL)
–10mg/dL
Decreased Ca2+ level
Parathyroid glands release PTH
Ca2+ level in blood increases
Osteoclasts release Ca2+ from bone
Calcium is reabsorbed from urine by the kidneys
Calcium absorption in the small intestine increases via vitamin D synthesis

Hormones in Calcium homeostasis

Hormone	***Effect on bones***	***Effect on GI***	***Effect on kidneys***
PTH ↑Ca^{++} , ↓PO_4 levels in blood	Supports osteoclast resorption	Indirect effects via ↑calcitriol	Supports Ca^{++} resorption and PO_4 excretion, activates 1-hydroxylation
Vitamin D ↑Ca^{++} , ↑PO_4 levels in blood	Bone resorption	↑Ca^{++} and PO_4 absorption	Increases Ca absorption
Calcitonin ↓Ca++ , ↓PO4 levels in blood when ↑ Ca++	Inhibits osteoclast resorption	No direct effects	Promotes Ca++ and PO4 excretion

Parathyroid hormone

- Parathyroid hormone (PTH) plays a key role in the **regulation of calcium and phosphate homeostasis and vitamin D metabolism.**
- **The four parathyroid glands** lie behind the lobes of the thyroid. The parathyroid chief cells respond directly to changes in calcium concentrations *When serum ionised calcium levels fall, PTH secretion rises.*

Role of Parathyroid hormone (PTH) in Calcium Homeostasis

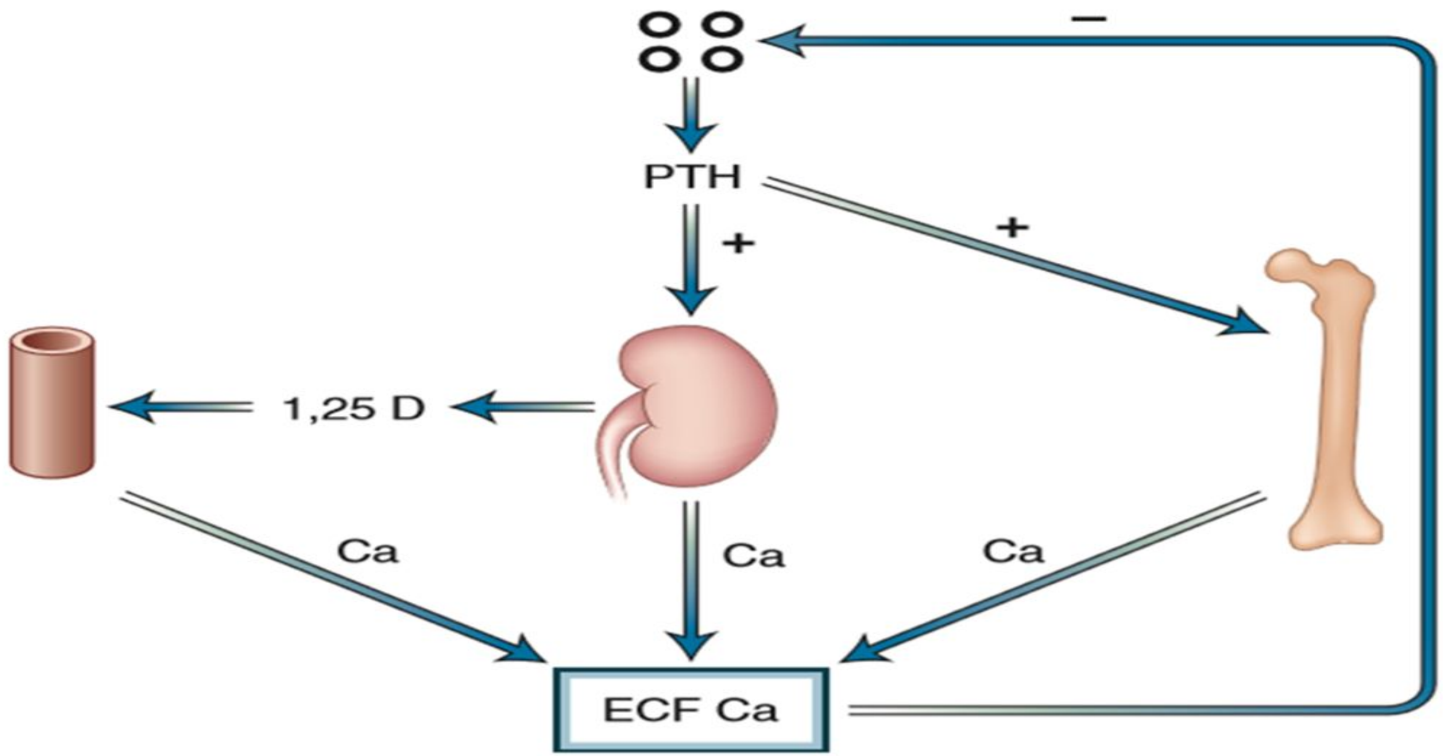


Figure 28-2 Parathyroid hormone (PTH)-calcium feedback loop that controls calcium homeostasis. Four organs—the parathyroid glands, intestine, kidney, and bone—together determine the parameters of calcium homeostasis. +, positive effect; –, negative effect; 1,25 D, 1,25-dihydroxyvitamin D_3; ECF, extracellular fluid.

William's textbook of Endocrinology 13th edition

Role of vitamin D in Calcium Metabolism

- The principal function of vitamin D is to **maintain adequate plasma calcium concentration.**

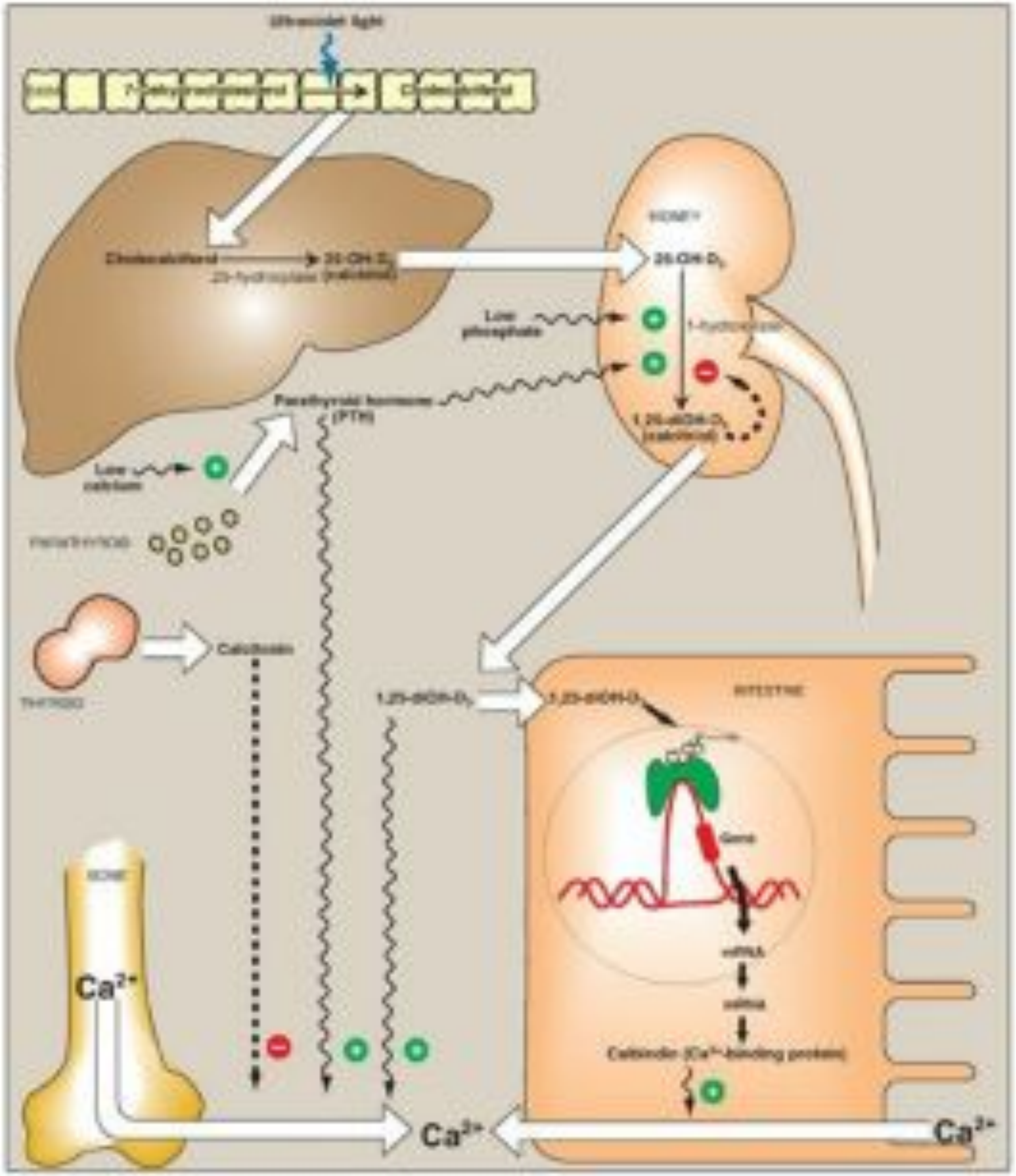
- Calcitriol achieves this in three ways:;
  - 1) it **increases intestinal absorption** of calcium
  - 2) it **reduces excretion of calcium** (by stimulating reabsorption in the distal renal tubules); and
  - 3) **stimulating resorption** (demineralization) of bone when blood calcium is low.

Regulation of 25-hydroxycholecalciferol 1-hydroxylase:

- 1,25-diOH-D3's formation is tightly regulated by the level of plasma phosphate and calcium ions.

- Low plasma calcium levels trigger the secretion of parathyroid hormone (PTH) from parathyroid gland.
- PTH upregulates the 1-hydroxylase. Thus, hypocalcemia caused by insufficient dietary calcium results in elevated levels of plasma 1,25-diOH-D3.

- Low plasma phosphate directly stimulates 1-hydroxylase activity .



CALCITONIN

- Calcitonin is a calcium regulating hormone.
- It is proved that calcitonin originates from special cells, called "C-cells", parafollicular cells.
- C-cells constitute an endocrine system which, are derived from "neural crest" and are found in **thyroid, parathyroids and in thymus.**
- Calcitonin is a single chain polypeptide, having a mol. wt. of 3600.
- It contains 32 amino acids.

Mechanism of Action

- 1. **Role of Cyclic AMP:** Calcitonin binds to specific calcitonin receptors on the plasma membrane of bone osteoclasts and renal tubular epithelial cells, activates adenyl cyclase which **increases c-AMP level** which mediates the cellular effects of the hormone.

- 2. **Cellular Shift:** Calcitonin may directly affect the relative distribution of bone cells and number of **osteoclasts decreased.**

- 3. **pH Change:** Calcitonin may regulate pH at cellular level producing **more alkaline medium which diminishes resorption**.

METABOLIC ROLE

- Calcitonin acts both on bone and kidneys account for:
  - **Hypocalcaemia**
  - **Hypophosphataemia**

- **Action on Bones:**
  - Calcitonin **inhibits the resorption of bones by osteoclasts** and thereby reduced mobilization of Ca and inorganic PO4 from bones into the blood.
  - It also stimulates influx of phosphates in bones.
  - It **increases osteoblasts cells**, which are thought to be involved in bone laying.

cont..

Action on Kidneys:

- The hormone acts on the distal tubule and ascending limb of Loop of Henle and **decreases tubular reabsorption of both calcium and inorganic PO4 t**hus producing calcinuria and phosphaturia.

- The hormone **inhibits α-1-hydroxylase and inhibits synthesis of 1,25-di-OH-D3** thus decreasing calcium absorption from intestine.

Both the above effects account for hypocalcaemia

CLINICAL ASPECT

- Abnormal calcitonin secretion is now proved in case of "**medullary carcinoma of thyroid**".
- Histologically, medullary carcinoma arises from para-follicular C-cells of thyroid.
- Frequently patients with this tumour have been found to have associated cutaneous neuromas, adrenal tumours and parathyroid enlargement.

THANK YOU